Machine Learning for the Quantified Self

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1 Introduction

Human Activity Recognition (HAR) encompasses methodologies aimed at classifying human actions based on sensor observations of movements and environmental context. There is a demand for accurate HAR in healthcare, elderly care, fitness tracking, and smart home applications.

In their seminal work, Anguita et al. (2013) [1] used integrated smartphone sensors—such as accelerometers and gyroscopes—to recognize daily living activities. They collected data from 30 volunteers performing six distinct activities: *standing*, *sitting*, *lying down*, *walking*, *walking upstairs*, and *walking downstairs*. Their approach yielded promising results, demonstrating the feasibility of smartphones for HAR research.

Our research expands on [1], aiming to address the following key research question:

- **Can modern smartphones, combined with contemporary machine learning techniques, reliably recognize human activities under diverse real-world conditions?**

We first replicated [1] , using their public dataset and classical machine learning methods. However, our models performed poorly when applied to data from a different sensor configuration, suggesting that sensor variation significantly affects HAR performance. To investigate this, we generated our own dataset using an iPhone 14 Pro and modern machine learning techniques, closely following [1].

2 Methodology

Two participants (us) performed the full activity protocol ten times each, resulting in forty total data recording sessions. Each protocol was executed twice per participant: once with the iPhone 14 Pro placed in the front-right pocket and once on the left side of the belt, allowing us to capture the effect of sensor placement on signal quality and model performance. Throughout every session, triaxial accelerometer and gyroscope data were collected at a sampling rate of

Label ID	Activity
1	Walking
2	Walking Upstairs
3	Walking Downstairs
4	Sitting
5	Standing
6	Lying Down

Table 1: Activity labels used for classification

Static Activities			**Dynamic Activities**		
No.	Activity	Time (sec)	No.	Activity	Time (sec)
0	Start (StandingPos)	0	7	Walk (1)	15
1	Stand (1)	15	8	Walk (2)	15
2	Sit (1)	15	9	Walk Downstairs (1)	12
3	Stand (2)	15	10	Walk Upstairs (2)	12
4	Lay Down (1)	15	11	Walk Downstairs (2)	12
5	Sit (2)	15	12	Walk Upstairs (2)	12
6	Lay Down (2)	15	13	Walk Downstairs (3)	12
			14	Walk Upstairs (3)	12
			15	Stop	0
Total Time: 192 sec					

Table 2: Protocol of activities for the HAR experiment.

50Hz using the iPhone's built-in sensors. The sequence and duration of all activities are detailed in Table 2, and the activity labels used for classification are summarized in Table 1.

Representative samples of the raw and calibrated sensor data are shown in Figures 1 and 2, respectively (see next page).

2.1 Experimental Setup

All sensor data was processed in Python (version 3.10) using NumPy, pandas, and scikit-learn for analysis and model development. Deep learning models were implemented with TensorFlow and Keras. Hyperparameter optimization was conducted using Optuna, and figures were created with Matplotlib and Seaborn.

3 Data Pre-processing

Prior to feature extraction and model training, all raw sensor data underwent a systematic pre-processing pipeline to improve data quality and ensure consistency across sessions.

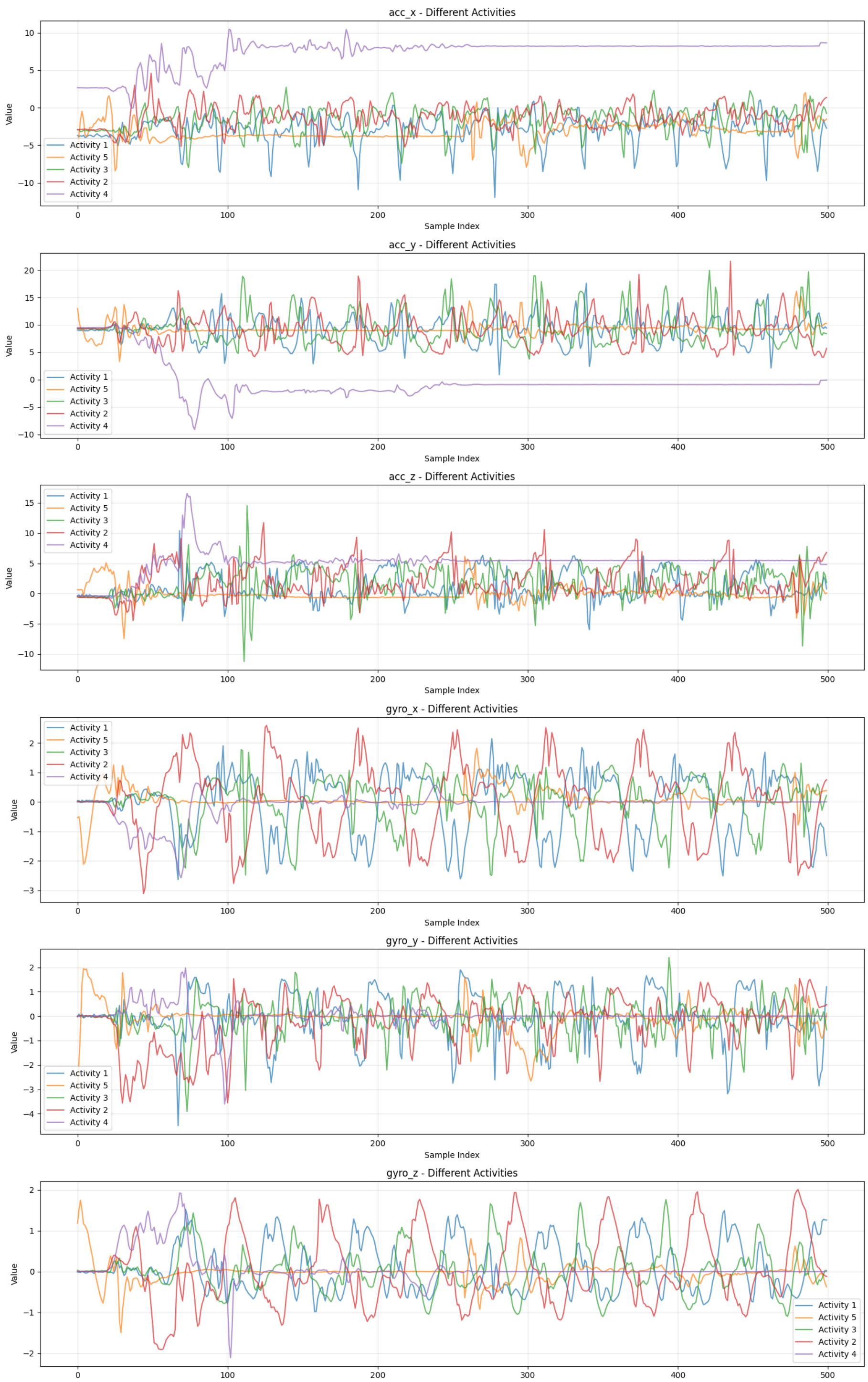


Figure 1: Raw sensor data by activity

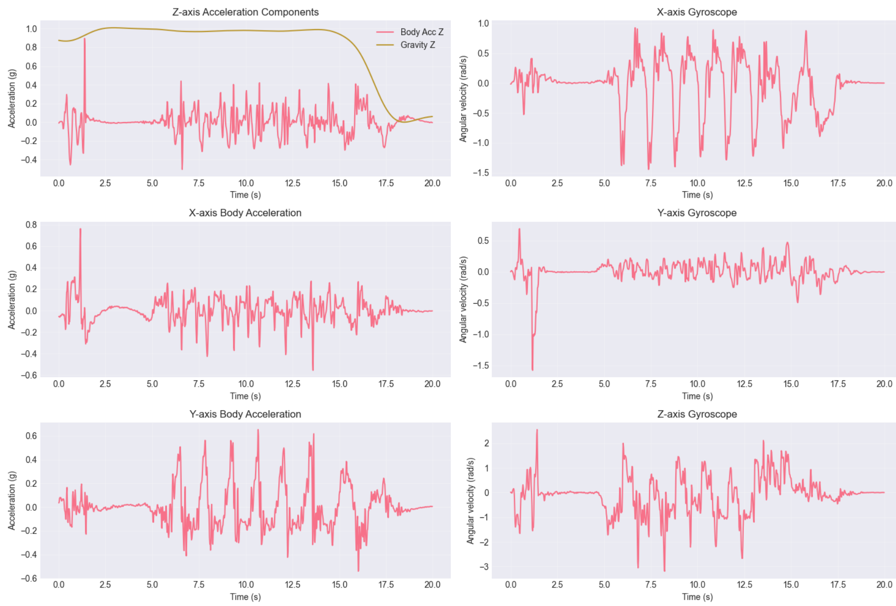


Figure 2: Calibrated Raw Signals

Attribute	Legacy (UCI)	Field (This Study)
Participants	30	2
Sensors	Acc/Gyro	Acc/Gyro
Activities	6	6
Duration	192 sec $\times$ 2	192 sec $\times$ 20
Placement	Waist	Pocket
Sampling Rate	50 Hz	50 Hz

Table 3: Summary of dataset characteristics

First, all accelerometer and gyroscope signals were resampled to a constant frequency of 50 Hz to account for any minor temporal misalignments during acquisition. Second, each sensor channel was processed using a median filter, followed by a third-order Butterworth low-pass filter with a cutoff frequency of 20 Hz. This two-step filtering approach effectively removed high-frequency noise while preserving the essential movement patterns required for activity recognition.

To further enhance the robustness of downstream feature extraction, the acceleration signal was separated into body acceleration and gravity components using an additional low-pass Butterworth filter at 0.3 Hz. This step allowed for independent analysis of gravitational effects and dynamic body motion, used for differentiating static and dynamic activities.

The denoised signals were then segmented into overlapping windows of fixed duration—specifically, 2.56-second windows with 50% overlap—yielding segments suitable for statistical analysis while maximizing the number of training instances. Within each window, signal normalization was applied to minimize the impact of inter-participant variability and to standardize the feature distribution for model input.

Finally, through annotation and pre-processing, we ensured that the data supplied to the feature engineering and modeling stages was clean and accurate. Each windowed segment was assigned an activity label in accordance with the experimental protocol. During data collection, annotation was performed manually using the stopwatch laps alongside the recordings. This allowed us to capture precise timestamps for each activity transition.

3.1 Windowing Rationale

The choice of a 2.56-second window with 50% overlap for segmenting the sensor data was directly informed by prior work in the field. This window length balances the need to capture sufficient temporal context for reliable activity recognition, while preserving a high number of training samples for statistical robustness. Empirical analyses in previous studies [1] [2] [3], have shown that a window size around 2.5 seconds is well-suited to the periodicity and duration of common activities such as walking or stair climbing, enabling the extraction of meaningful temporal and frequency-domain features. The use of overlapping windows further enhances model performance by increasing the diversity of training examples and smoothing label transitions across activity boundaries.

4 Feature Set

The construction of the feature set in this study is grounded in the extraction and transformation of raw triaxial accelerometer and gyroscope signals, denoted as tAcc-XYZ and tGyro-XYZ, respectively. All sensor signals were sampled at a constant rate of 50 Hz and subjected to a rigorous preprocessing pipeline to improve data quality and reduce noise. Specifically, a median filter was first applied, followed by a third-order low-pass Butterworth filter with a corner frequency of 20 Hz, effectively attenuating high-frequency noise artifacts.

After initial filtering, the acceleration data were further decomposed into body and gravity components (tBodyAcc-XYZ and tGravityAcc-XYZ) via an additional low-pass Butterworth filter with a 0.3 Hz cutoff. This decomposition is essential for separating the effects of gravity from dynamic movements, thereby enabling a more nuanced interpretation of human activity.

To further enrich the signal representations, several derived variables were calculated:

- **Jerk signals**: The temporal derivatives of both the body acceleration and gyroscope signals were computed to form jerk signals (tBodyAccJerk-XYZ, tBodyGyroJerk-XYZ), which capture abrupt changes in motion.

- **Magnitude signals**: For each three-dimensional signal, the Euclidean norm was calculated, resulting in variables such as tBodyAccMag, tGravityAccMag, tBodyAccJerkMag, tBodyGyroMag, and tBodyGyroJerkMag. These features summarize overall movement intensity regardless of direction.

- **Frequency domain signals**: Selected signals were transformed into the frequency domain using the Fast Fourier Transform (FFT), yielding variables prefixed with 'f', such as fBodyAcc-XYZ, fBodyGyro-XYZ, and their magnitude and jerk variants. These features are intended to capture periodicities and spectral characteristics associated with different activities.

Feature	**Description**
`mean()`	Mean value
`std()`	Standard deviation
`mad()`	Median absolute deviation
`max()`	Maximum value
`min()`	Minimum value
`sma()`	Signal magnitude area
`energy()`	Sum of squared values divided by the number of values
`iqr()`	Interquartile range
`entropy()`	Signal entropy
`arCoeff()`	Autoregressive coefficients (Burg method, order 4)
`correlation()`	Correlation coefficient between signal pairs
`maxInds()`	Index of frequency component with the largest magnitude
`meanFreq()`	Weighted average of frequency components
`skewness()`	Skewness of the frequency domain signal
`kurtosis()`	Kurtosis of the frequency domain signal
`bandsEnergy()`	Energy in specified FFT frequency bands
`angle()`	Angle between averaged signal vectors

Table 4: Statistical features used in HAR classification

A comprehensive set of statistical descriptors was then computed from these signals to form the final feature vectors. Table 4 provides an overview of the extracted features, each designed to capture a distinct aspect of the underlying sensor signals, from basic statistics (mean, standard deviation) to more advanced measures (entropy, autoregressive coefficients, frequency-domain statistics).

The influence of these features on classification performance is visualized in Figure 3, which presents the top 20 most important features as determined by model-based feature importance analysis. In the original dataset, `body_acc_x_max`, `body_acc_x_std`, and `body_acc_x_energy` emerge as particularly influential.

By contrast, the field data reveals a more uniform feature importance profile, with `acc_z_median`, `gyro_x_fft_max`, and `corr_acc_y_acc_z` among the top

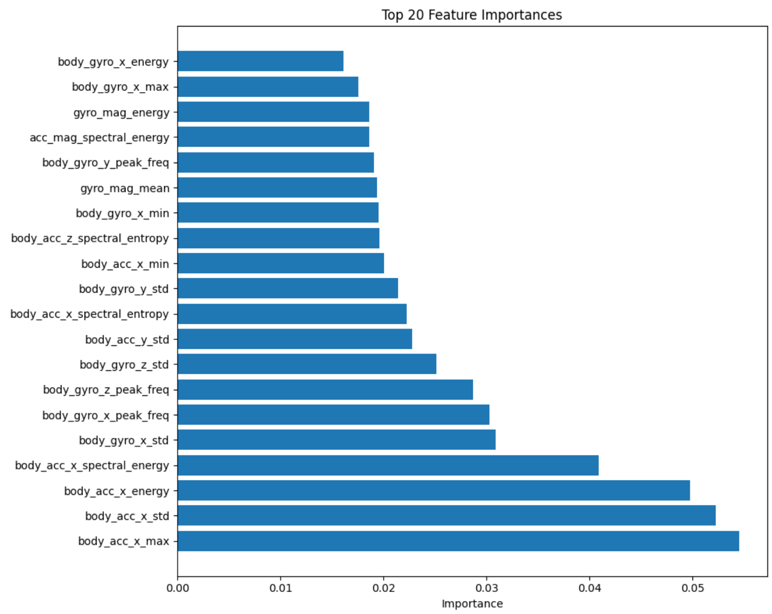


Figure 3: Top 20 features by importance in the original dataset

features, suggesting that the discriminative value of specific features is highly dataset-dependent.

5 Model Selection

A range of machine learning and deep learning algorithms were evaluated. We benchmarked classical models against more recent neural network architectures to assess their relative strengths.

We benchmarked Support Vector Machines (SVM), Random Forests (RF), and Gradient Boosting Machines (GBM). These models were selected due to their established performance in structured, tabular datasets and their historical success in HAR literature. The SVM was configured for multiclass classification using a one-vs-one scheme, while RF and GBM leveraged ensemble learning to capture complex non-linearities within the feature space.

The benchmark was contrasted against several deep learning models. In particular a Convolutional Neural Network (CNN), a Long Short-Term Memory network (LSTM), a hybrid CNN-LSTM, and a Temporal Convolutional Network (TCN). The CNN and CNN-LSTM models are capable of learning spatial and temporal dependencies directly from raw or minimally-processed sequences, while the TCN is specifically designed to handle long-range temporal dependencies using causal and dilated convolutions.

Each model was trained and evaluated using identical train-test splits to en-

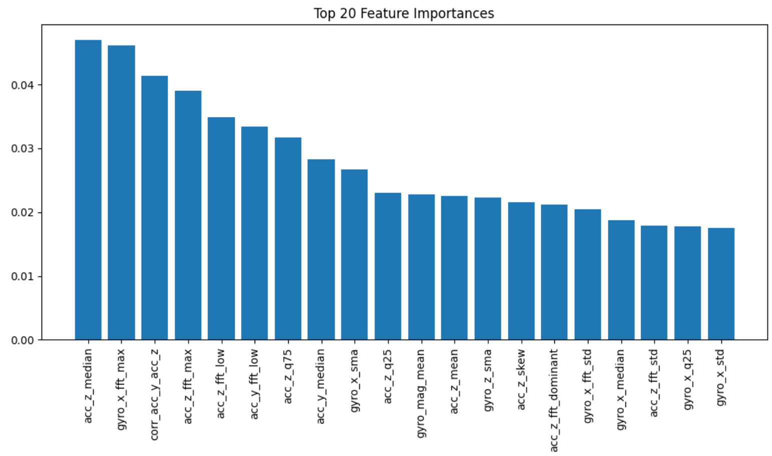


Figure 4: Top 20 features by importance in the field dataset

sure comparability. Performance was assessed primarily using overall accuracy, confusion matrices, and per-class recall, with an emphasis on real-world generalizability to field data. The results (see Results sections) provide a comprehensive comparison of classifier performance.

6 Hyperparameter Tuning

Optimal model performance is highly dependent on the careful selection of hyperparameters. For classical machine learning models, hyperparameters such as the number of trees (RF), learning rate and tree depth (GBM), and the kernel and regularization parameter (SVM) were tuned via grid search or randomized search, using cross-validation on the training set to avoid overfitting.

For deep learning models, a more extensive hyperparameter optimization process was employed. We used the Optuna optimization framework to automate the search over model-specific hyperparameters, including learning rate, batch size, number of convolutional filters, kernel size, dilation rates, number of layers, dropout rates, and sequence length (for windowed input). Early stopping and learning rate reduction strategies were incorporated to further guard against overfitting and to reduce computational time.

For the Temporal Convolutional Network (TCN) we used number of filters, kernel size, dilation rates, dropout rate, and skip connections or weight normalization were systematically varied. The optimization objective was to maximize validation accuracy on held-out data. The final model configuration was selected based on its performance on the validation set and then retrained on the full training data before final testing.

All hyperparameter choices, as well as the search space, are documented in ‘Assignment.ipynb‘.

7 Results

Detailed evaluation results are available in the code repository in `Assignment.ipynb`.

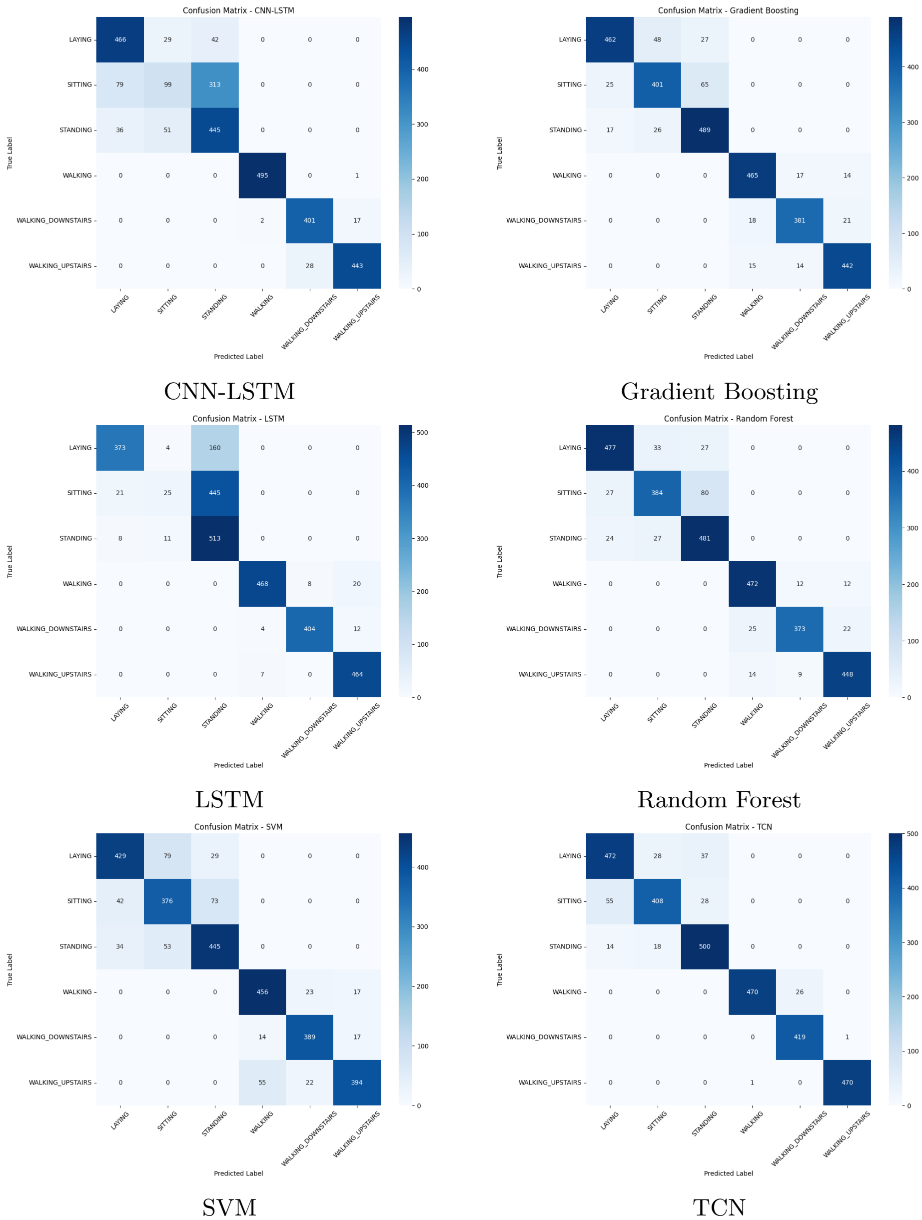


Figure 5: Confusion matrices for six classification models

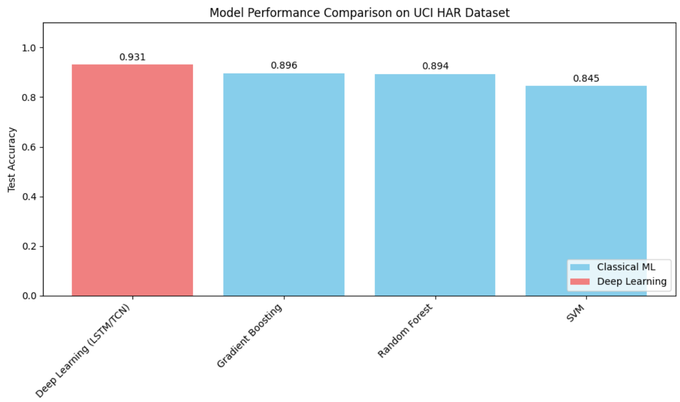
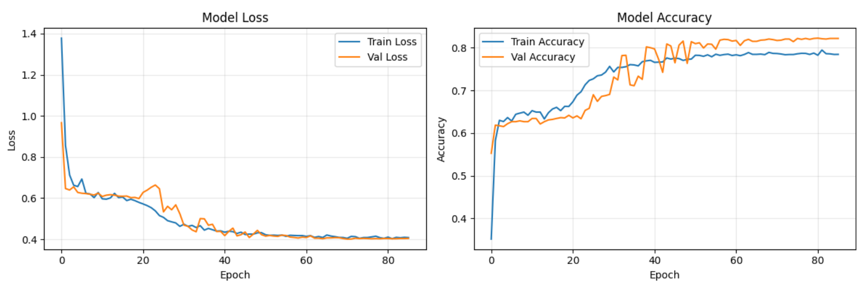
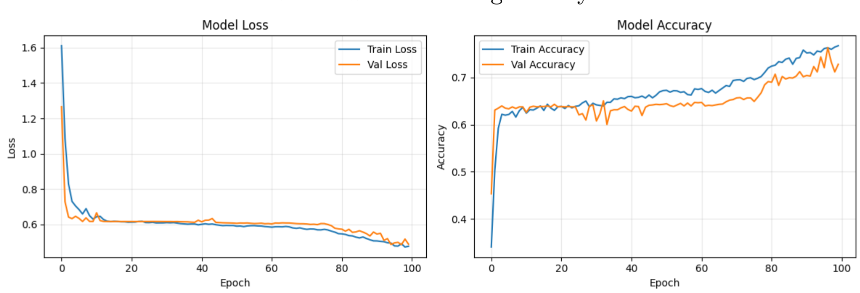


Figure 6: Performance comparison of models on legacy dataset



CNN-LSTM Training History



LSTM Training History

Figure 7: Training accuracy and loss over epochs for CNN-LSTM and LSTM models

8 Discussion and Analysis

Analysis of the results revealed notable differences in model performance between the controlled (legacy) dataset and the newly collected field data. Classical models such as Support Vector Machines and Random Forests demonstrated

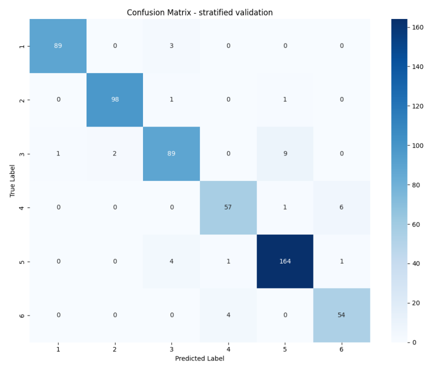


Figure 8: field_data_validation

high accuracy on the legacy dataset, largely due to the well-structured nature of the original data and the effectiveness of engineered statistical features. Deep learning models, particularly CNN-LSTM and Temporal Convolutional Networks, performed competitively and showed a greater capacity to model temporal dependencies in sequential sensor data, offering improved robustness when applied to less constrained, real-world conditions.

While this study provides valuable insights into the challenges and performance of smartphone-based HAR, several limitations must be acknowledged. First, the dataset comprises only two participants, which restricts the generalizability of the findings to broader populations. In terms of temporal coverage, all data were collected within a single day, limiting our ability to assess model stability across multiple days or varying routines. Environmental diversity was also constrained: all activities were performed outdoors at Amstel station, where a stairway with approximately 18 steps was selected to closely match the conditions described in the original study (see Appendix for a photograph of the stairway). While this location enabled realistic data collection for stair ascent and descent at the required cadence, the lack of varied indoor and outdoor environments means that the results may not fully reflect the complexities encountered in daily life across diverse settings.

Future work should seek to expand the temporal and environmental scope of the data collection—recording activities on multiple days and in a broader range of locations—to enhance the robustness and ecological validity of smartphone-based HAR systems.

References

[1] Davide Anguita et al. "A public domain dataset for human activity recognition using smartphones." In: *Esann*. Vol. 3. 1. 2013, pp. 3–4.

[2] Chiraz BenAbdelkader, Ross Cutler, and Larry Davis. “Stride and cadence as a biometric in automatic person identification and verification”. In: *Proceedings of Fifth IEEE international conference on automatic face gesture recognition*. IEEE. 2002, pp. 372–377.

[3] Jhun-Ying Yang, Jeen-Shing Wang, and Yen-Ping Chen. “Using acceleration measurements for activity recognition: An effective learning algorithm for constructing neural classifiers”. In: *Pattern recognition letters* 29.16 (2008), pp. 2213–2220.

Appendix: Stairway Used for Data Collection

Figure 9: Photograph of the stairway at Amstel station used for stair ascent and descent activities.